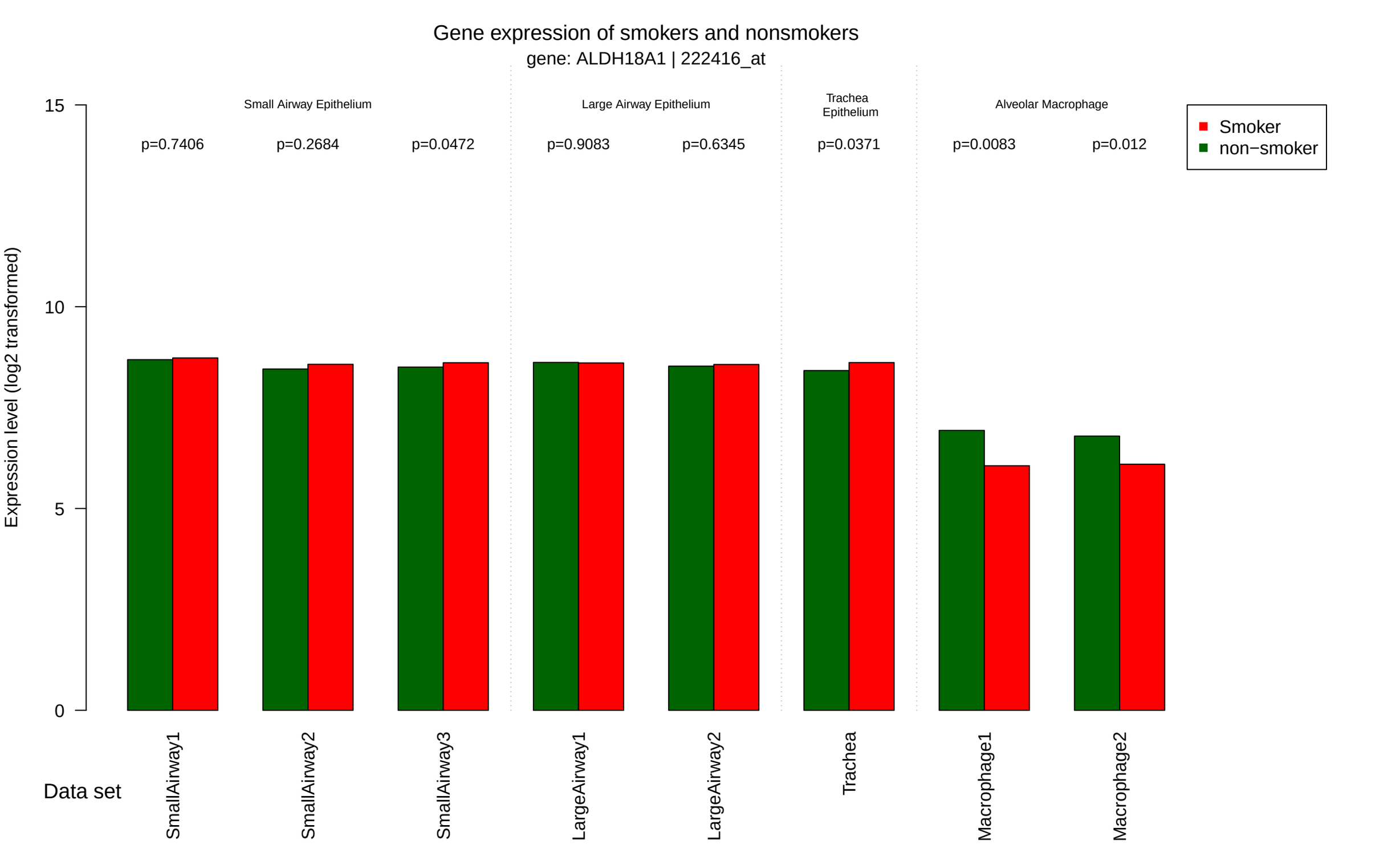
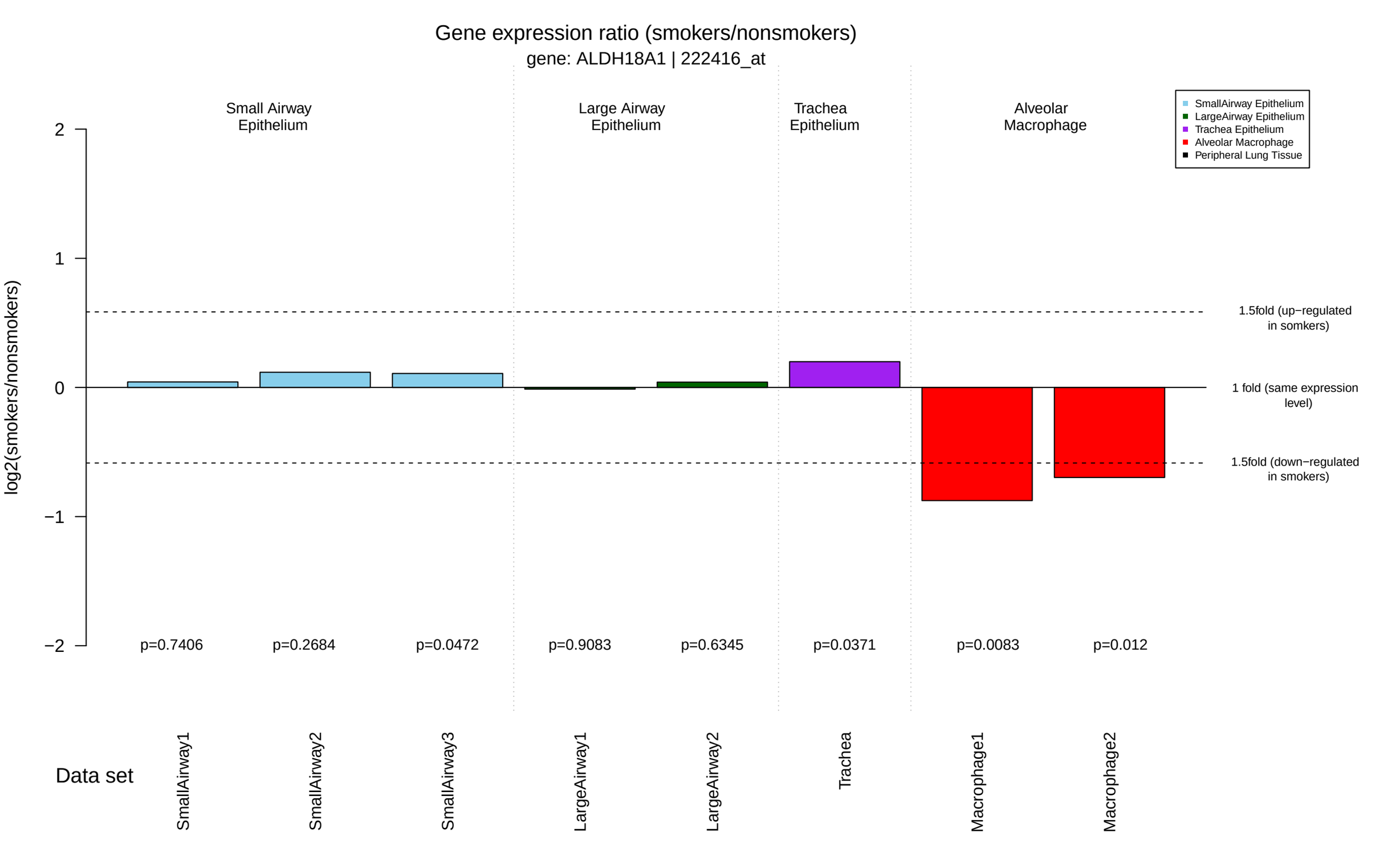
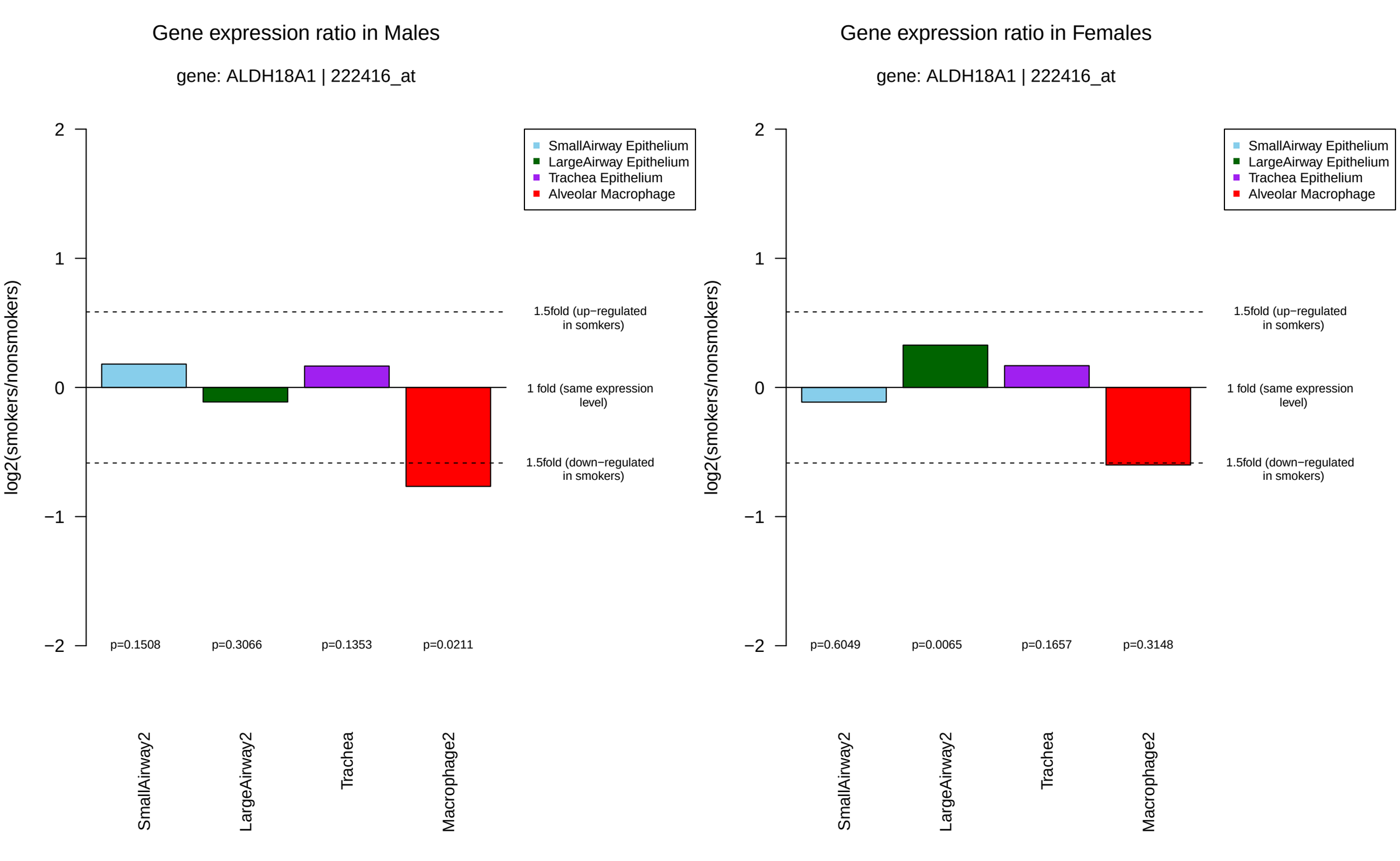
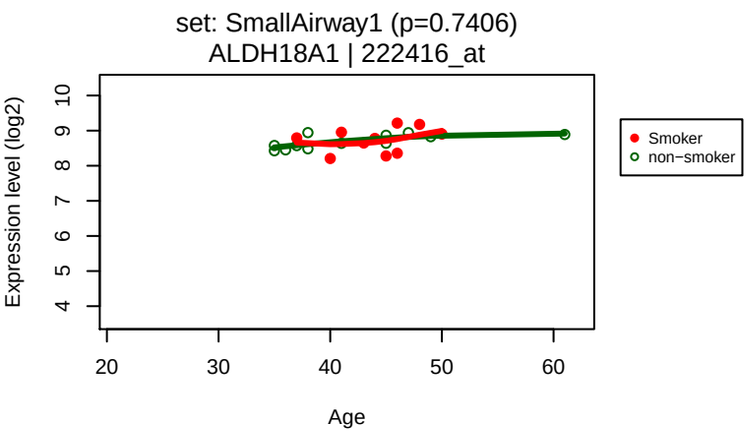
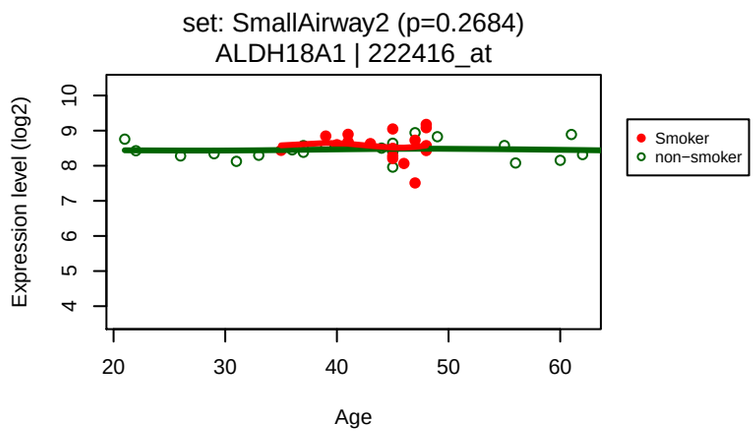
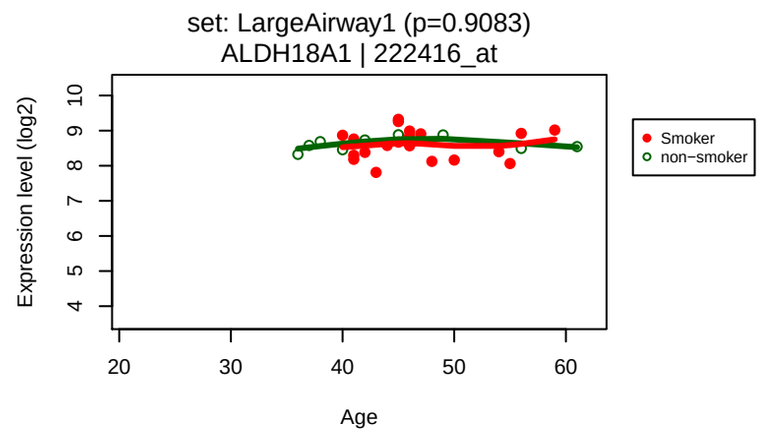

Gene expression of smokers and nonsmokers
gene: ALDH18A1 | 222416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7406
p=0.2684
p=0.0472
p=0.9083
p=0.6345
p=0.0371
p=0.0083
p=0.012
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

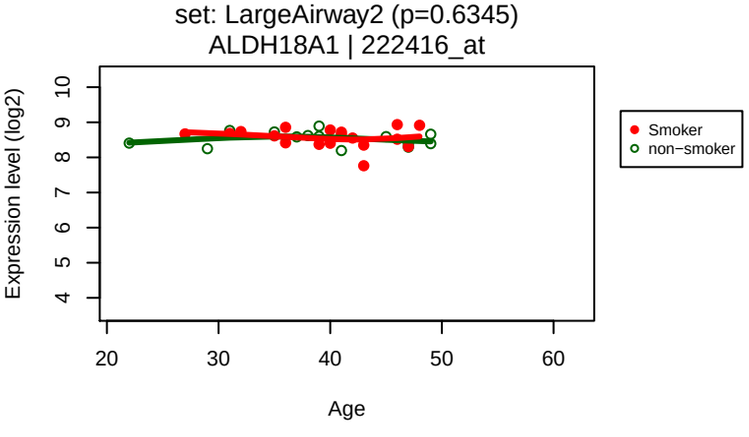

Gene expression ratio (smokers/nonsmokers)
gene: ALDH18A1 | 222416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7406
p=0.2684
p=0.0472
p=0.9083
p=0.6345
p=0.0371
p=0.0083
p=0.012
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

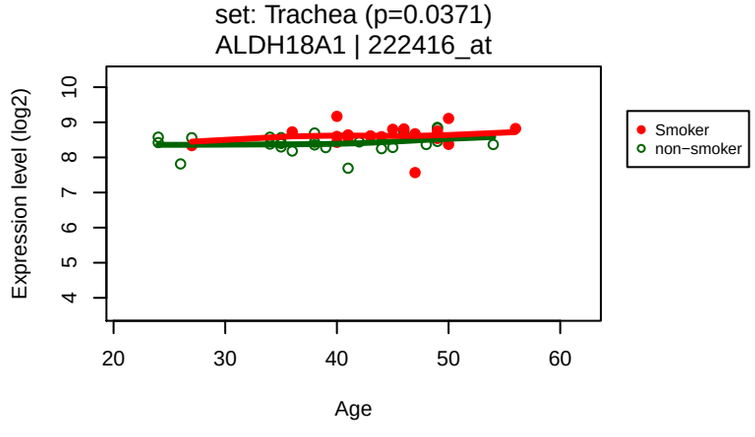

Gene expression ratio in Males
gene: ALDH18A1 | 222416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1508
p=0.3066
p=0.1353
p=0.0211
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH18A1 | 222416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6049
p=0.0065
p=0.1657
p=0.3148
SmallAirway2
LargeAirway2
Trachea
Macrophage2

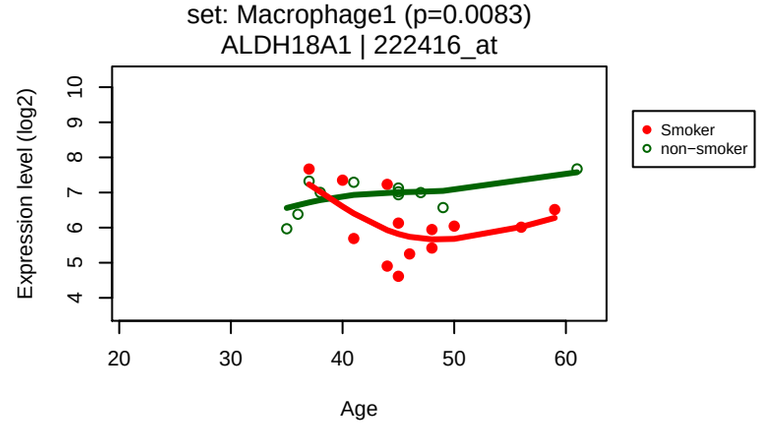

set: SmallAirway1 (p=0.7406)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker

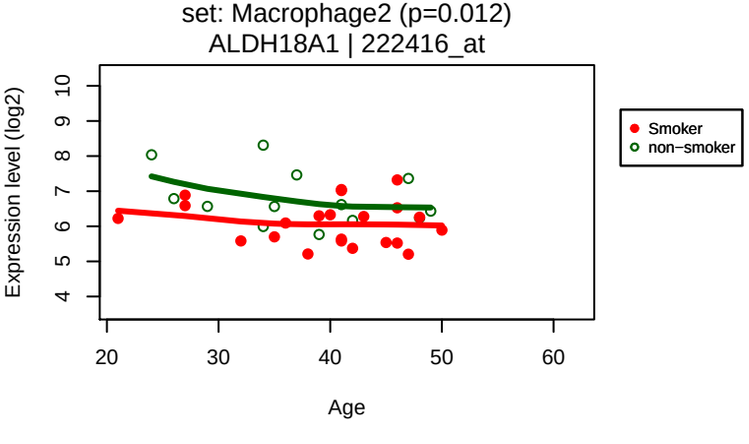

set: SmallAirway2 (p=0.2684)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker

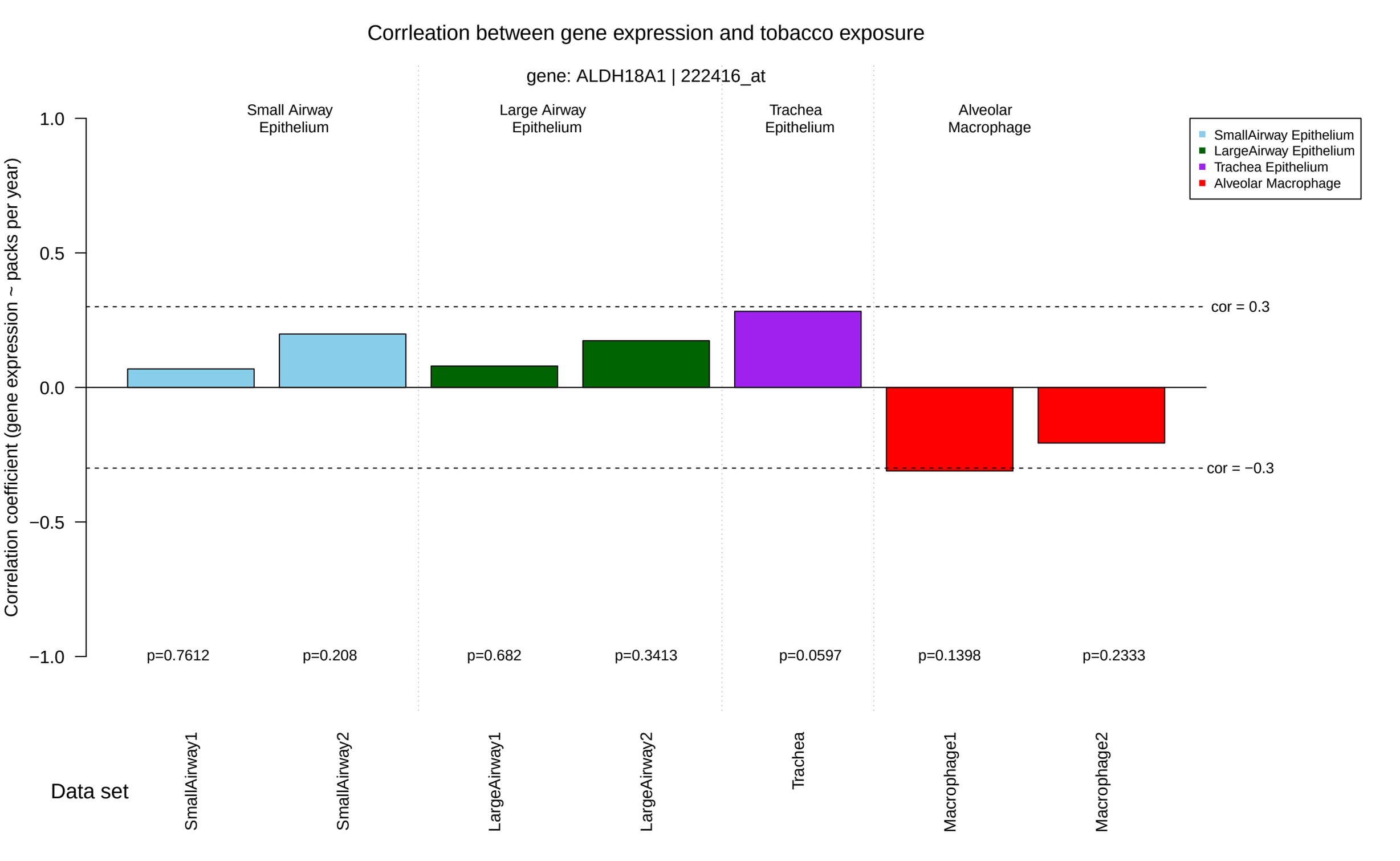

set: LargeAirway1 (p=0.9083)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6345)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0371)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0083)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.012)
ALDH18A1 | 222416_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH18A1 | 222416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7612
p=0.208
p=0.682
p=0.3413
p=0.0597
p=0.1398
p=0.2333
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2